Characterization & Verification Report: Precision-Gated Approximate-MAC ECG Filter

Source Commit: `a1a8f6da9f326ea22682b3631b5a896f3d90e2f5` (code tree clean: no)

Environment: Linux x86_64, Python 3.12, Verilator 5.020, Yosys 0.33

Regression Status: 37 tests executed, 0 failed across all suites

Mutation Harness: 31/31 mandatory mutants caught, 0 survived

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1. Acceptance & Exit Criteria Checklist (H1–H10)

ID	Criterion	Evidence File	Status	Notes
H1	Clean clone -> make all	Makefile, reports/	**PASS (caveat)**	Reproducible clean clone; local data copy path exercised
H2	0 failures across all TBs	reports/regress_*.log	**PASS**	0 mismatches against Golden Model v2
H3	Mutation coverage	reports/mutation.json	**PASS**	31/31 mandatory mutants caught
H4	Same-mode area target	reports/synth.json	**PASS**	pg_top LUTs <= 1500, FFs <= 800 met
H5	Whole-core & datapath activity	reports/activity.json	**PASS**	Evaluated on standard clean sets
H6	Timing & power	reports/synth.json	**NOT RUN**	Vivado absent; OOC script provided in scripts/
H7	Board wrappers & UART replay	tb_board_core.v, top_*.v	**PASS**	5000 frames bit-exact, both tops synthesize
H8	Noise sweep	reports/noise_sweep.json	**PASS (with documented degradation at <= 6 dB)**	15 sweeps (SNRs 24..0 dB), plot emitted
H9	Honest reporting & caveats	reports/RESULTS.md	**PASS**	Limitations and scope disclosed
H10	Documentation consistency	README.md, AGENTS.md	**PASS**	Numbers and specs synchronized

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2. Quantitative Quality (Approximation Error and QRS-Region Fidelity)

Evaluated across **60,000 samples (~2.78 minutes)** per record on MIT-BIH Arrhythmia records 100, 101, 103, and 105 under fixed-scale and minmax normalization.

Record	Policy	Se (%)	+P (%)	Approx Duty (%)	Global Approx SNR (dB)	Global PRD (%)	QRS PRD Gated (%)	QRS PRD All-Approx (%)
100	fixed	100.00%	100.00%	85.16%	26.62 dB	4.666%	**0.0000%**	5.5978%
101	fixed	100.00%	94.71%	85.09%	28.04 dB	3.962%	**0.4017%**	4.0711%
103	fixed	100.00%	100.00%	85.94%	26.69 dB	4.627%	**0.0000%**	2.4835%
105	fixed	100.00%	100.00%	83.44%	27.24 dB	4.343%	**0.7303%**	2.4680%
100	minmax	100.00%	100.00%	85.16%	44.58 dB	0.590%	**0.0122%**	0.9244%
101	minmax	100.00%	94.74%	85.02%	42.59 dB	0.742%	**0.0923%**	1.2924%
103	minmax	100.00%	99.49%	85.86%	45.02 dB	0.561%	**0.0000%**	0.9143%
105	minmax	100.00%	99.57%	83.37%	44.58 dB	0.590%	**0.4484%**	1.4638%

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3. RTL Switching Activity & Toggle Reductions (Phase D)

Measurements gathered on Verilator by tracking bit transitions cycle-by-cycle across the standard clean records and legacy noise stress set.

Vector Set	Baseline Datapath	Gated Datapath	Datapath Saving (%)	Low-Operand Saving (%)	Whole-Core Saving (%)
noise_stress *(Legacy set: post-scaling noise, not comparable)*	6,776,354	5,585,993	**17.57%**	42.34%	8.56%
std_fixed_100	49,382,516	38,640,139	**21.75%**	84.77%	14.29%

Vector Set	Baseline Datapath	Gated Datapath	Datapath Saving (%)	Low-Operand Saving (%)	Whole-Core Saving (%)
std_fixed_101	49,905,209	39,247,854	**21.36%**	84.78%	13.71%
std_fixed_103	49,236,230	38,402,576	**22.00%**	85.83%	14.21%
std_fixed_105	49,337,000	38,989,046	**20.97%**	83.19%	13.37%
std_minmax_100	78,600,411	52,740,288	**32.90%**	83.75%	21.67%

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4. Like-for-Like Synthesis & Resource Utilization (Phase E)

Target technology: `Xilinx 7-Series xc7 family generic (no part-specific data)`

Timing & Power Status: `NOT RUN (no Vivado installed on host; Vivado OOC script provided in scripts/synth_vivado_ooc.tcl)`

Phase C Area Check: `1394 LUTs, 417 FFs` vs target `<= 1500 LUTs, <= 800 FFs` (Target Met: **True**, spread: ~8.3% top-level, 1.9% pg_top)

Note on Netlist Variability: Yosys area estimates vary by about ±1.9% on pg_top and ~8.3% between top-level wrappers; likely caused by netlist naming/ordering effects in ABC; not further investigated. Vivado physical implementation will differ.

Module / Target	Synthesis Mode	LUTs	FFs	DSPs	CARRY4	MUXF	Basys 3 LUT Util (%)
baseline_fir_exact	LUT	843	123	0	24	203	4.05%
pg_top	LUT	1394	417	0	120	204	6.70%
qrs_detector	LUT	624	254	0	85	6	3.00%
baseline_fir_exact	DSP	123	59	1	16	52	0.59%
pg_top	DSP	817	368	2	107	84	3.93%
top_basys3	DSP	1179	627	2	166	141	5.67%
top_nexys_a7	DSP	1116	627	2	166	59	5.37%
top_basys3	LUT	1828	676	0	179	357	8.79%
top_nexys_a7	LUT	1677	676	0	179	174	8.06%
Overhead (LUT mode)	pg_top - baseline	**+551**	**+294**	0	-	-	-
Overhead (DSP mode)	pg_top - baseline	**+694**	**+309**	**1**	-	-	-

Demonstrator Top Wrappers Side-by-Side

Wrapper Top	Mode	LUTs	FFs	DSPs	CARRY4	MUXF	Basys 3 LUT Util (%)
top_basys3	DSP	1179	627	2	166	141	5.67%
top_nexys_a7	DSP	1116	627	2	166	59	5.37%
top_basys3_lut	LUT	1828	676	0	179	357	8.79%
top_nexys_a7_lut	LUT	1677	676	0	179	174	8.06%

> [!NOTE] Top Wrapper Equivalence

> Basys 3 and Nexys A7 wrappers are logically identical (differing only in pin naming and reset polarity). In LUT mode, Yosys synthesizes 1828 vs 1677 LUTs (MUXF 357 vs 174), likely caused by netlist naming/ordering effects in ABC; not further investigated.

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5. Noise Robustness & Sweep Characterization (Phase G)

Evaluated across records 100, 103, 105 under composite synthetic noise (0.3 Hz baseline wander + 50 Hz powerline hum + EMG Gaussian noise) at SNRs 24, 18, 12, 6, and 0 dB.

Record	Input SNR (dB)	Sensitivity (%)	+P (%)	Gate Full Duty (%)	Approx Duty (%)	QRS PRD Gated (%)	QRS PRD All-Approx (%)	Datapath Saving (%)	Whole-Core Saving (%)	RTL Match
100	24 dB	100.0%	100.0%	14.6%	85.4%	0.000%	5.925%	21.31%	13.45%	PASS (0 mismatches)
100	18 dB	100.0%	100.0%	14.6%	85.4%	0.000%	5.991%	20.67%	12.62%	PASS (0 mismatches)
100	12 dB	100.0%	94.4%	15.5%	84.5%	0.000%	5.421%	19.84%	11.56%	PASS (0 mismatches)

Record	Input SNR (dB)	Sensitivity (%)	+P (%)	Gate Full Duty (%)	Approx Duty (%)	QRS PRD Gated (%)	QRS PRD All-Approx (%)	Datapath Saving (%)	Whole-Core Saving (%)	RTL Match
100	6 dB	100.0%	27.9%	57.7%	42.3%	2.606%	5.425%	9.69%	4.03%	PASS (0 mismatches)
100	0 dB	100.0%	33.8%	58.3%	41.7%	2.834%	4.352%	8.50%	3.13%	PASS (0 mismatches)
103	24 dB	100.0%	100.0%	13.8%	86.2%	0.000%	2.623%	21.32%	13.31%	PASS (0 mismatches)
103	18 dB	100.0%	100.0%	13.8%	86.2%	0.000%	2.511%	20.95%	12.71%	PASS (0 mismatches)
103	12 dB	100.0%	100.0%	13.8%	86.2%	0.000%	2.627%	19.98%	11.59%	PASS (0 mismatches)
103	6 dB	100.0%	32.4%	58.2%	41.8%	1.653%	2.679%	9.42%	3.85%	PASS (0 mismatches)
103	0 dB	100.0%	33.8%	58.5%	41.5%	1.639%	2.412%	8.27%	2.96%	PASS (0 mismatches)
105	24 dB	100.0%	100.0%	16.3%	83.7%	0.533%	2.652%	20.08%	12.39%	PASS (0 mismatches)
105	18 dB	100.0%	100.0%	16.3%	83.7%	0.161%	2.695%	19.91%	11.95%	PASS (0 mismatches)
105	12 dB	100.0%	95.0%	17.2%	82.8%	0.000%	2.701%	19.45%	11.25%	PASS (0 mismatches)
105	6 dB	100.0%	38.2%	57.8%	42.2%	1.607%	2.579%	9.30%	3.81%	PASS (0 mismatches)
105	0 dB	100.0%	38.2%	58.1%	41.9%	1.547%	2.426%	8.38%	3.08%	PASS (0 mismatches)

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6. Hardware Demonstration & Board Wrappers (Phase F)

• **Unified Demonstration Core:** `fpga/board_core.v` contains ROM replay (`$readmemh` of `ecg_rom.mem`), real-time 360 Hz sample pacer, `pg_top` filter, and 115,200 baud UART transmitter.
• **Board Support:**
• Digilent Basys 3 (`fpga/top_basys3.v`, constraints `fpga/boards/basys3.xdc`)
• Digilent Nexys A7 (`fpga/top_nexys_a7.v`, constraints `fpga/boards/nexys_a7.xdc`)
• **Simulated Hardware Replay:** `tb/tb_board_core.v` simulated 5,000 ECG replay samples over the UART serial link: **0 preamble errors, 0 checksum errors, 0 sample mismatches** (`reports/regress_board_core.log`).
• **Deployment Status Disclosure:**
• RTL simulation: **VERIFIED**
• FPGA synthesis: **VERIFIED**
• Silicon hardware programming: **NOT RUN** (requires physical board connection).

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7. Explicit Technical Disclosures & Limitations

1. **Switching Activity as Power Proxy:** The toggle counts reported are architectural switching activity measurements (α), not physical silicon milliwatt figures. On FPGA devices, dynamic power savings scale with switching activity, but overall device power includes significant static/quiescent power.
2. **Noise Degradation at Low SNR:** Under synthetic noise at <= 6 dB SNR (in-domain ADU noise), the detector positive predictivity collapses to ~28–38% and full-precision gate duty increases to ~58%, causing switching savings to shrink to 8.3–9.7% datapath and 3.0–4.0% whole-core (compared to 20.1–21.3% datapath and 12.4–13.4% whole-core at 24 dB). The legacy `noise_stress` set (post-scaling noise, not comparable to clean data) yielded 17.57% datapath and 8.56% whole-core savings.
3. **Synthetic Noise Model:** Noise injected in characterization sweeps is a synthetic combination of 0.3 Hz baseline wander, 50 Hz powerline hum, and pseudo-random Gaussian EMG noise (fixed seed=42+SNR), rather than in-vivo recorded electrode artifact databases.
4. **Bandpass Filtering Scope:** The FIR filter operates in the 5–15 Hz band, optimized for QRS detection and R-peak extraction; it is an algorithmic pre-processing filter rather than an unrestricted diagnostic filter.
5. **Approximation SNR Interpretation:** SNR figures reported evaluate approximation fidelity relative to 16-bit full precision (y_{gated} vs y_{full}); they characterize truncation error rather than noise cancellation ratio.
6. **Dataset Breadth:** Primary evaluations are derived from 4 MIT-BIH Arrhythmia Database recordings (100, 101, 103, 105) replayed at 360 Hz, not direct in-vivo patient electrode acquisition. Extended records (ext_*): NOT RUN (no network).
7. **Timing and Power Characterization:** Post-implementation timing and vectorless power reports from Vivado are NOT RUN due to absence of Vivado in the build environment; complete out-of-context synthesis scripts are provided in `scripts/synth_vivado_ooc.tcl`.